Winning Space Race with Data Science

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9 March 2026

GitHub Repository:
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone

Outline

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Executive Summary

Project Objective

The goal of this project is to predict whether the **Falcon 9 first stage booster** will land successfully.

Why This Matters

- Falcon 9 reusability significantly reduces launch costs.
- Predicting landing success improves mission planning and cost estimation.
- Provides business insight into launch reliability patterns.

Methodologies Used

- Data collection via SpaceX REST API and Web Scraping
- Data wrangling and preprocessing
- Exploratory Data Analysis (EDA)
- Interactive analytics (Folium + Plotly Dash)
- Classification modeling (Logistic Regression, SVM, Decision Tree, KNN)

Final Outcome

- Built and evaluated multiple classification models.
- Identified the best-performing model based on accuracy.
- Delivered interactive visual analytics dashboard.

Introduction

Background

SpaceX developed the **Falcon 9** rocket with reusable first-stage boosters to reduce launch costs.

The first stage:

1. Detaches after launch
2. Attempts controlled landing
3. Can be reused if landing is successful

Business Problem

If we can predict:

"**Will the booster land successfully?**"

Then:

1. Cost projections improve
2. Risk assessment becomes data-driven
3. Competitive intelligence improves

Project Question :

Can we build a machine learning model that predicts Falcon 9 first-stage landing success using historical launch data?

Section 1

Methodology

Methodology

Data Collection

1. Extracted launch data using SpaceX REST API
2. Scraped historical launch records from Wikipedia
3. Consolidated structured launch dataset

Data Wrangling

1. Cleaned missing values
2. Removed irrelevant columns
3. Converted categorical variables
4. Created binary target variable (Landing Success: 0/1)

Exploratory Data Analysis (EDA)

1. Payload vs Launch Site analysis
2. Orbit type vs Success rate
3. Flight number trends
4. Yearly launch success trends

Interactive Visual Analytics

1. Built geographic launch analysis using Folium
2. Developed interactive dashboard using Plotly Dash

Predictive Analysis

1. Built classification models:
    - Logistic Regression
    - Support Vector Machine (SVM)
    - Decision Tree
    - K-Nearest Neighbors (KNN)
2. Applied hyperparameter tuning using GridSearchCV
3. Evaluated models using accuracy and confusion matrix

Data Collection

- **Objective**

To gather structured and reliable launch data for predictive modeling.

- Data Sources

- **SpaceX REST API**
  - Launch metadata
  - Booster versions
  - Payload mass
  - Orbit type
  - Launch site
  - Landing outcome
- **Wikipedia Web Scraping**
  - Historical launch records
  - Launch dates
  - Booster information
  - Mission outcomes
  - Output Dataset

Final dataset included:

- Flight Number
- Launch Site
- Payload Mass
- Orbit Type
- Booster Version
- Landing Outcome (Target Variable)
- Repository Notebooks:
- SpaceX API Data Collection Notebook
- Web Scraping Notebook

Data Collection – SpaceX API

Data Source

Launch data was collected using the public SpaceX REST API.

API Endpoint
https://api.spacexdata.com/v4/launches

Data Collection Process

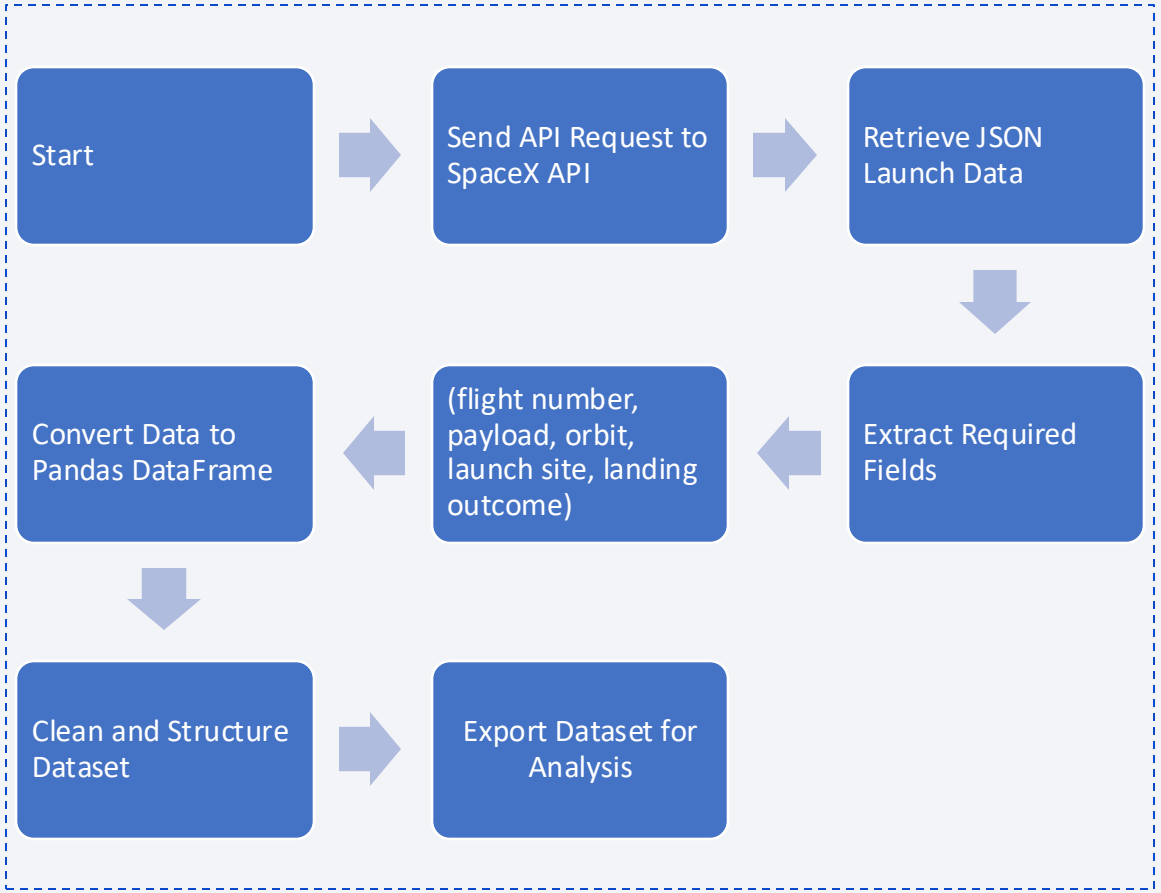
- Sent API request to retrieve launch records
- Received response in JSON format
- Parsed nested JSON objects to extract relevant features
- Selected important fields such as flight number, payload mass, orbit, launch site, and landing outcome
- Converted extracted data into a structured Pandas DataFrame
- Cleaned and stored the dataset for further analysis

Key Features Extracted

- Flight Number
- Launch Site
- Payload Mass (kg)
- Orbit Type
- Booster Version
- Landing Outcome

Notebook Reference

SpaceX API Data Collection Notebook
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

Data Source

Historical Falcon 9 launch records were collected by scraping launch tables from Wikipedia.

Source Page
https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Web Scraping Process

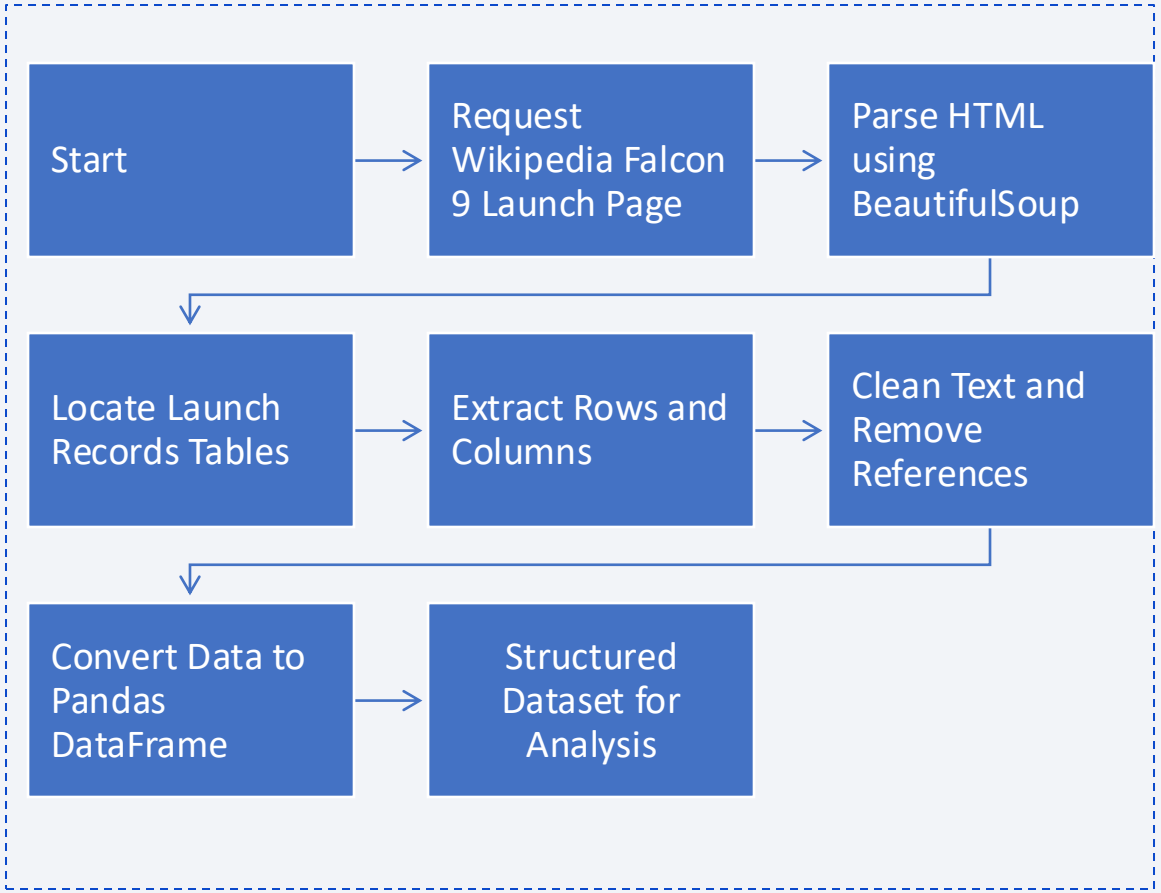
- Sent request to retrieve the Falcon 9 launch webpage
- Parsed HTML content using BeautifulSoup
- Identified tables containing launch records
- Extracted rows and columns containing mission details
- Cleaned extracted text and removed reference symbols
- Converted scraped data into a structured Pandas DataFrame

Data Extracted

- Launch Date
- Launch Site
- Payload Name
- Orbit Type
- Mission Outcome
- Booster Landing Result

GitHub Notebook Reference

Web Scraping Notebook:
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/jupyter-labs-webscraping.ipynb



Data Wrangling

Data Wrangling Process

After collecting the data, preprocessing steps were performed to prepare the dataset for analysis and modeling.

Key Data Processing Steps

- Loaded the collected launch dataset into a Pandas DataFrame
- Inspected dataset structure and identified missing values
- Removed unnecessary or redundant columns
- Handled missing values in payload mass and booster information
- Encoded categorical variables into usable formats
- Created a binary **landing outcome variable** (Success = 1, Failure = 0)

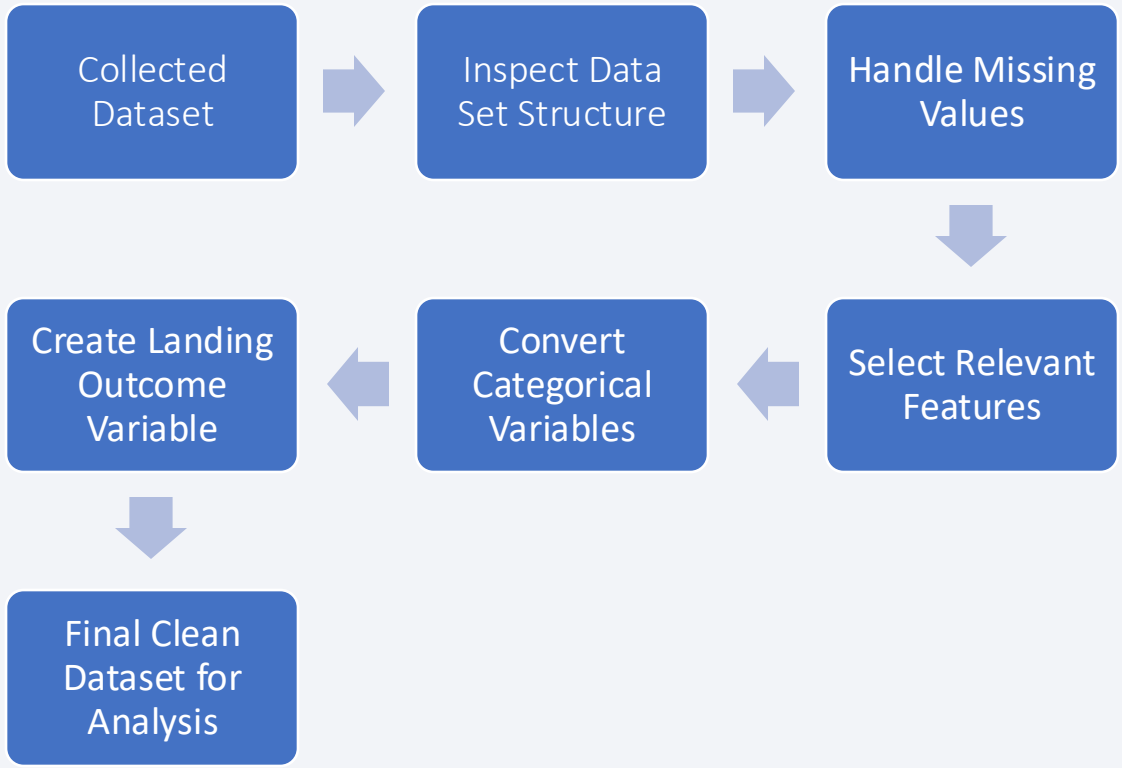
Output :

A cleaned and structured dataset ready for:

- Exploratory Data Analysis
- Interactive visualization
- Machine learning modeling

GitHub Reference

- Data Wrangling Notebook
  https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/labs-jupyter-spacex-Data_wrangling.ipynb



EDA with Data Visualization

EDA was performed to explore relationships between launch variables and landing success.

Key Visualizations Created

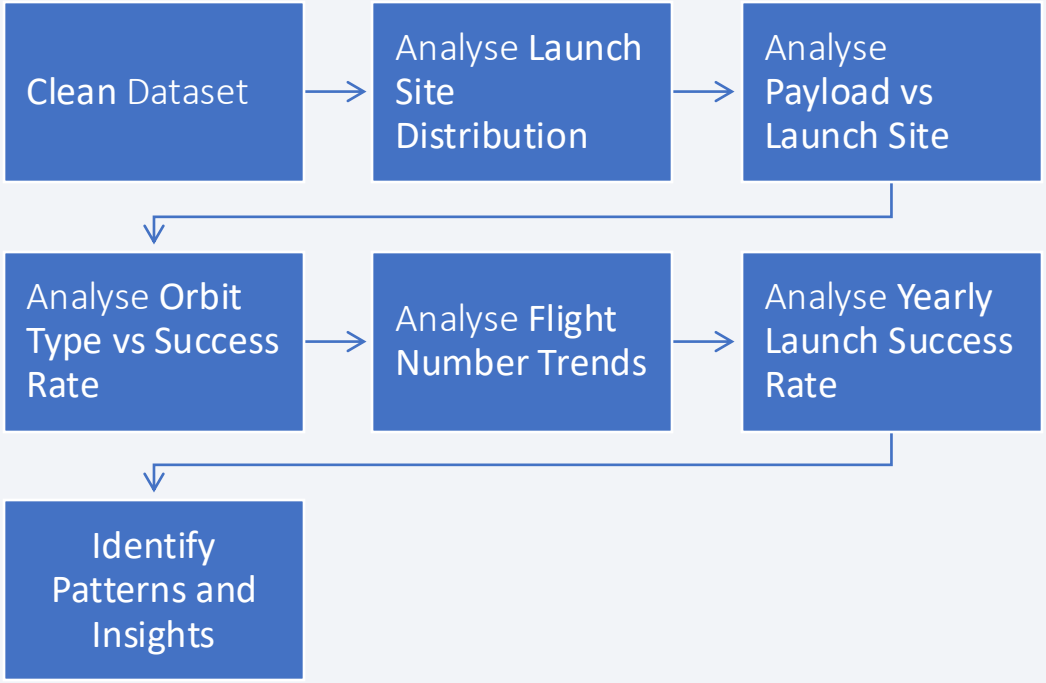
- **Scatter plot of Flight Number vs Launch Site**
- **Scatter plot of Payload Mass vs Launch Site**
- **Bar chart showing Success Rate by Orbit Type**
- **Scatter plot of Flight Number vs Orbit Type**
- **Scatter plot of Payload vs Orbit Type**
- **Line chart showing Yearly Launch Success Trends**

Purpose of Visualization

- Identify patterns affecting landing success
- Understand relationships between payload, orbit, and launch sites
- Discover historical success trends

GitHub Reference

EDA with Visualization Notebook
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/edadataviz.ipynb



EDA with SQL

SQL queries were used to analyze launch data and extract key insights from the dataset.

SQL Analysis Performed

Retrieved all **unique launch site names**

- **Identified launch sites beginning with "CCA"**
- **Calculated total payload carried by NASA missions**
- **Calculated average payload mass for booster version F9 v1.1**
- **Identified first successful ground landing date**
- **Retrieved boosters with successful drone ship landing with payload between 4000–6000 kg**
- **Calculated total number of successful and failed landing outcomes**
- **Identified boosters carrying the maximum payload mass**

Purpose of SQL Analysis

- Perform structured data exploration
- Extract mission statistics
- Identify performance trends in booster landings

GitHub Reference

EDA with SQL Notebook: https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Interactive Visual Analytics with Folium

Interactive maps were created to visualize **geographical launch locations and landing outcomes**.

Map Objects Added

- **Markers** to display the location of each launch site
- **Color-coded markers** to indicate launch success or failure
- **Circle markers** to highlight launch site coordinates
- **Lines** to measure distance between launch sites and nearby infrastructure
- **Distance labels** showing proximity to cities, highways, railways, and coastline

Purpose of Map Objects

- Identify geographic distribution of launch sites
- Visualize success and failure outcomes geographically
- Analyze proximity of launch sites to infrastructure and urban areas
- Understand location-based factors influencing launch success

Insights from Interactive Map

- Launch sites are generally located **close to coastlines** for safety reasons
- Launch sites are typically **far from large cities**
- Some sites are located near transportation infrastructure

GitHub Notebook Reference

Interactive Map Notebook
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Interactive Dashboard using Plotly Dash

An interactive dashboard was built to explore launch outcomes dynamically.

Visualizations Added

- **Pie Chart – Launch Success Count by Site**
  Shows total successful launches for each launch site.
- **Pie Chart – Site-specific Success Ratio**
  Displays success vs failure distribution for a selected launch site.
- **Scatter Plot – Payload Mass vs Launch Outcome**
  Visualizes relationship between payload mass and landing success.
- **Payload Range Slider**
  Allows filtering launches by payload mass range.

Purpose of Dashboard

- Enable interactive exploration of launch data
- Compare success rates across launch sites
- Identify payload ranges with higher landing success
- Allow users to dynamically filter and analyze data

GitHub Notebook Reference

Plotly Dash Dashboard Notebook
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/spacex-dash-app.py

Predictive Analysis (Classification)

Classification models were developed to predict whether the **Falcon 9 first stage booster will land successfully**.

Classification Models Built

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)

Model Evaluation

Models were evaluated using:

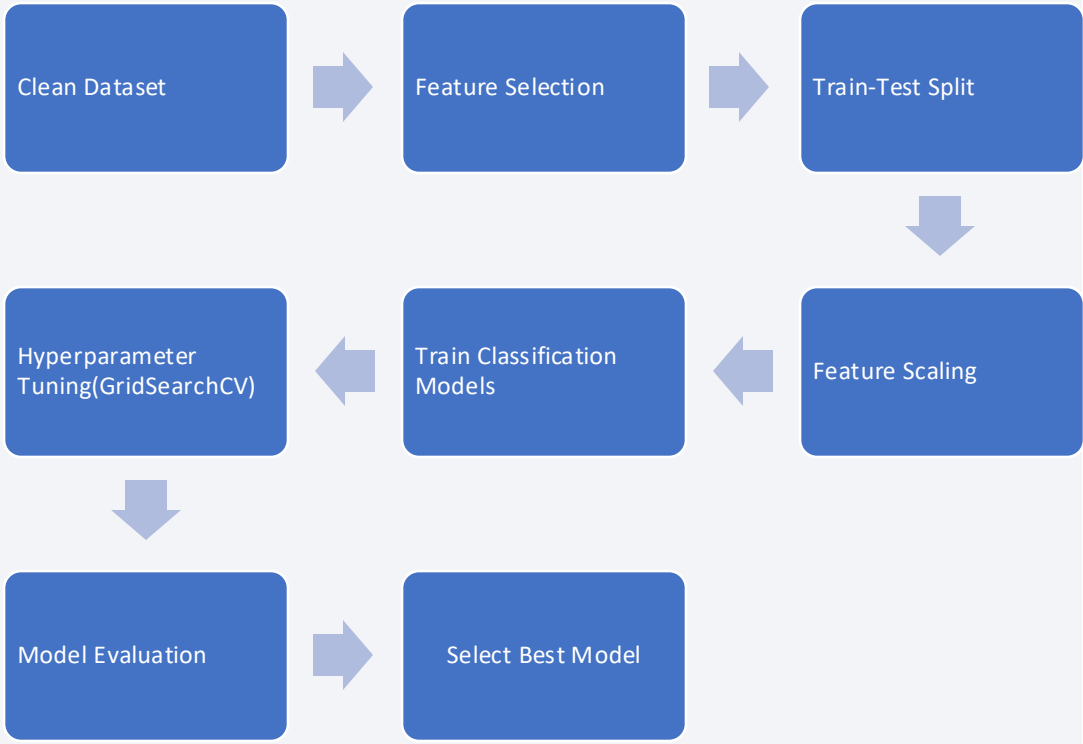
- Accuracy score
- Confusion matrix
- Cross-validation

Outcome

The model with the **highest accuracy after hyperparameter tuning** was selected as the best-performing classifier for predicting Falcon 9 landing success.

GitHub Notebook Reference

Predictive Modeling Notebook
https://github.com/Vidhi-Pandey04/Applied-Data-Science-Capstone/SpaceX_Machine_Learning_Prediction.ipynb



Results

Exploratory Data Analysis

- Identified relationships between **payload mass, orbit type, launch site, and landing success**.
- Observed that **launch success rates improved over time** as Falcon 9 technology matured.
- Certain **launch sites and orbit types showed higher success rates**.

Interactive Analytics

- **Folium Map:** Visualized launch site locations and landing outcomes geographically.
- **Plotly Dash Dashboard:** Interactive charts for launch success by site and payload vs outcome analysis.

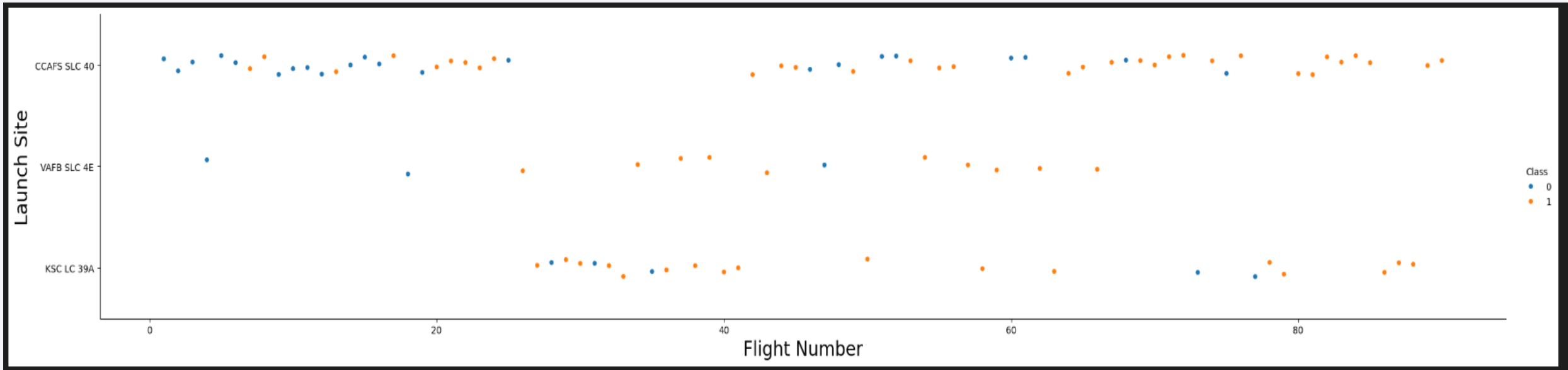
Predictive Analysis

- Built multiple classification models: **Logistic Regression, SVM, Decision Tree, and KNN**.
- Models evaluated using **accuracy and confusion matrix**.
- Decision Tree was the **best-performing model based on highest prediction accuracy**.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



Conclusion

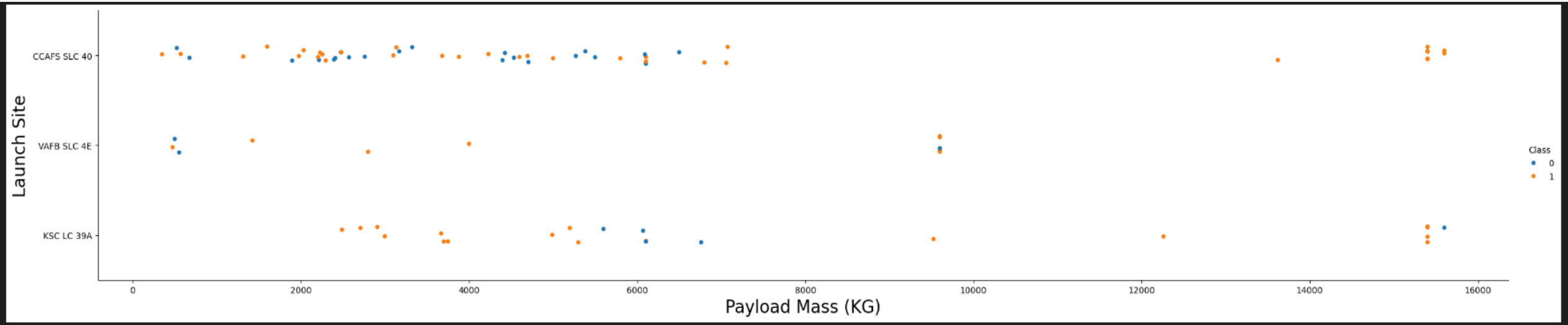
Early launches (low flight numbers) show more landing failures.

As flight numbers increase, the success rate improves significantly.

This indicates SpaceX improved booster landing technology over time.

Most launches occurred at CCAFS SLC-40 and KSC LC-39A, which also show higher success rates.

Payload vs. Launch Site



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Conclusion

Most successful landings occur for payload masses between 2000–6000 kg.

VAFB-SLC launch site rarely handles very heavy payloads (>10,000 kg).

Launch sites such as CCAFS SLC-40 and KSC LC-39A handle a wider range of payloads.

Higher payload masses generally increase landing difficulty, leading to some failures.

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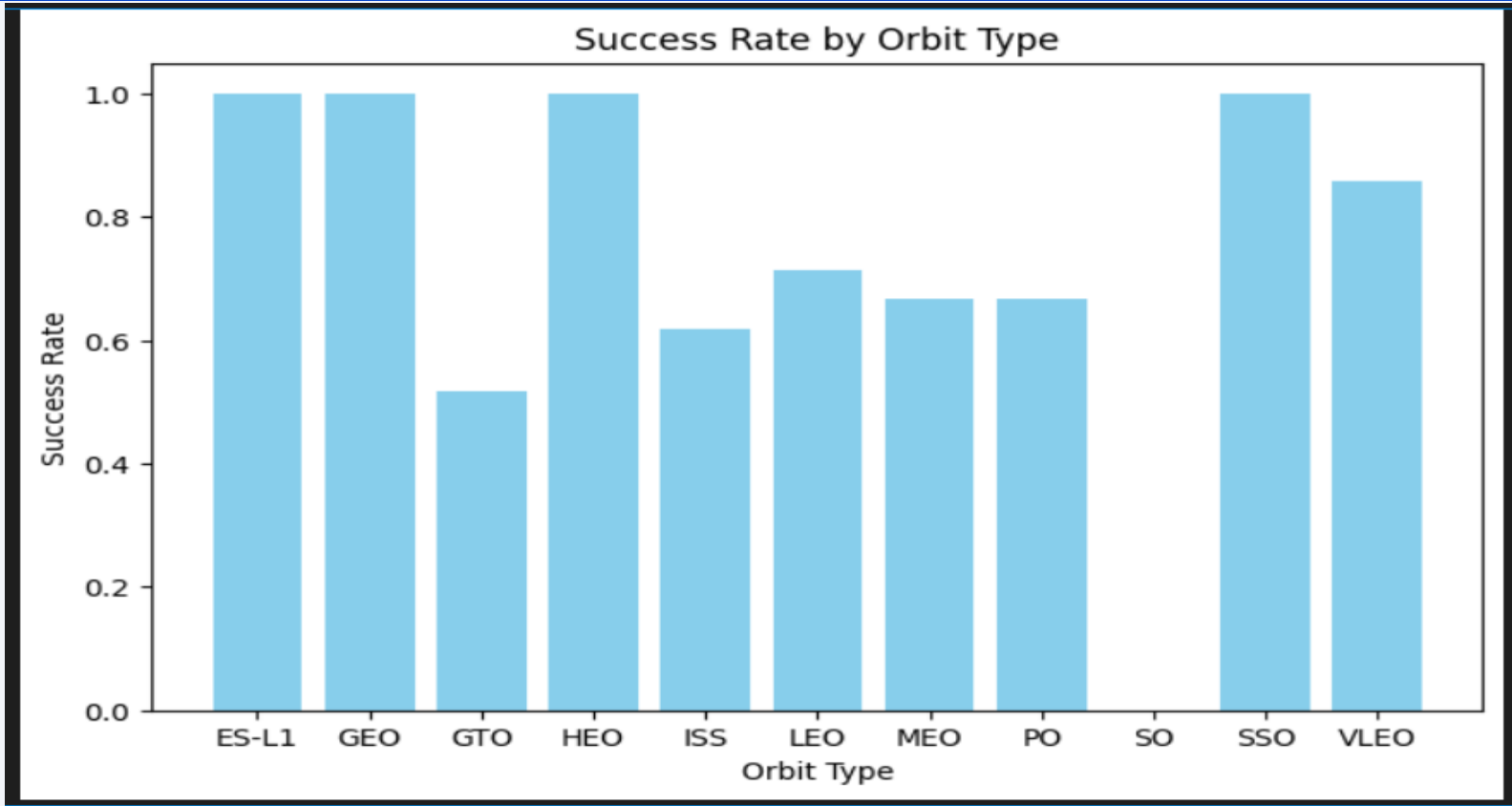
Success Rate vs. Orbit Type

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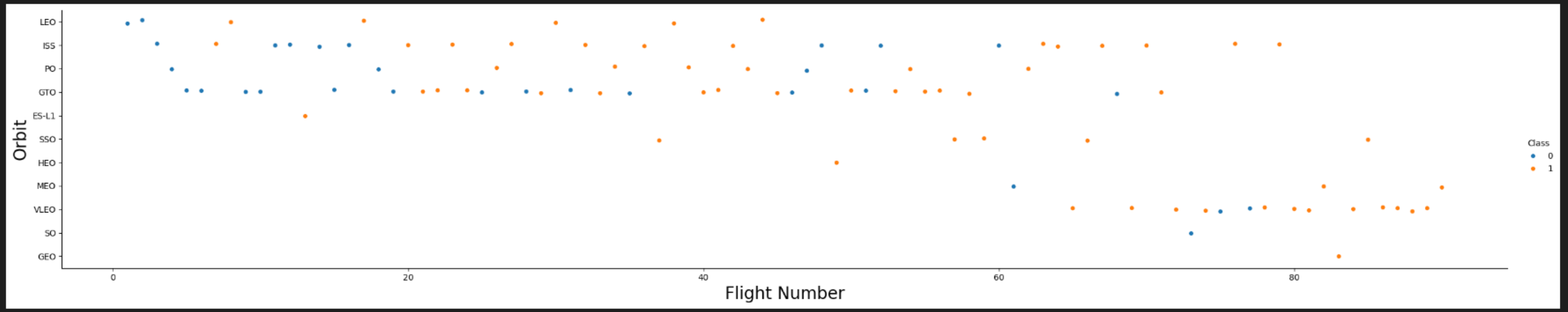
Conclusion

LEO orbit shows the highest success rate among missions.
ISS and Polar orbits also demonstrate relatively strong landing success rates.
GTO orbit has lower success rates, likely due to heavier payloads and higher energy requirements.
```

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Conclusion:
In LEO missions, landing success improves as flight numbers increase, showing technological improvement over time.
For GTO missions, success and failure occur across many flight numbers, indicating orbit complexity affects landing success.
Overall, experience from repeated launches contributes to higher success rates.

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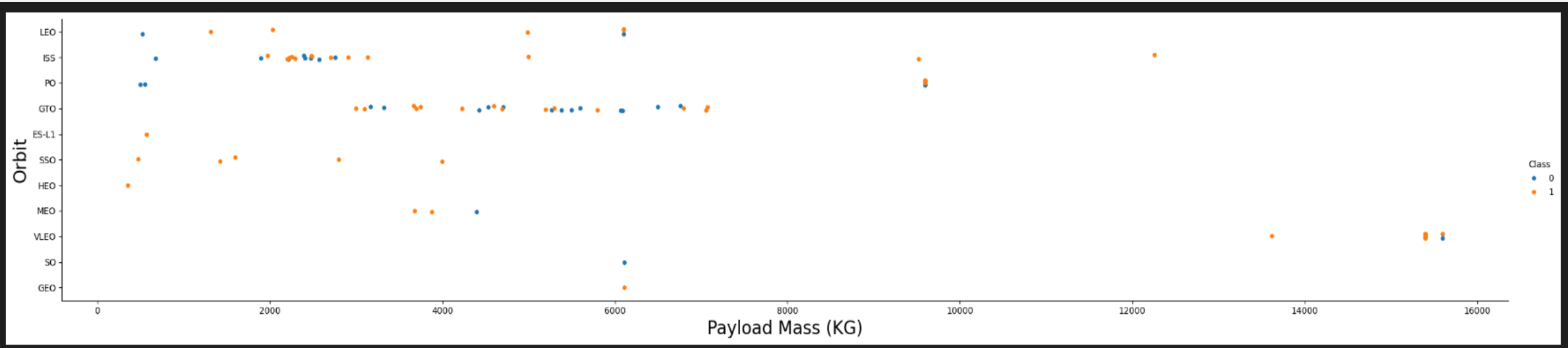
Payload vs. Orbit Type




```
With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.
However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.
Conclusion:

-Higher payload masses are more common in GTO missions, which often show mixed success outcomes.
-LEO, ISS, and Polar missions with moderate payload masses show higher landing success rates.
-This suggests payload weight and mission orbit both influence landing probability.
```

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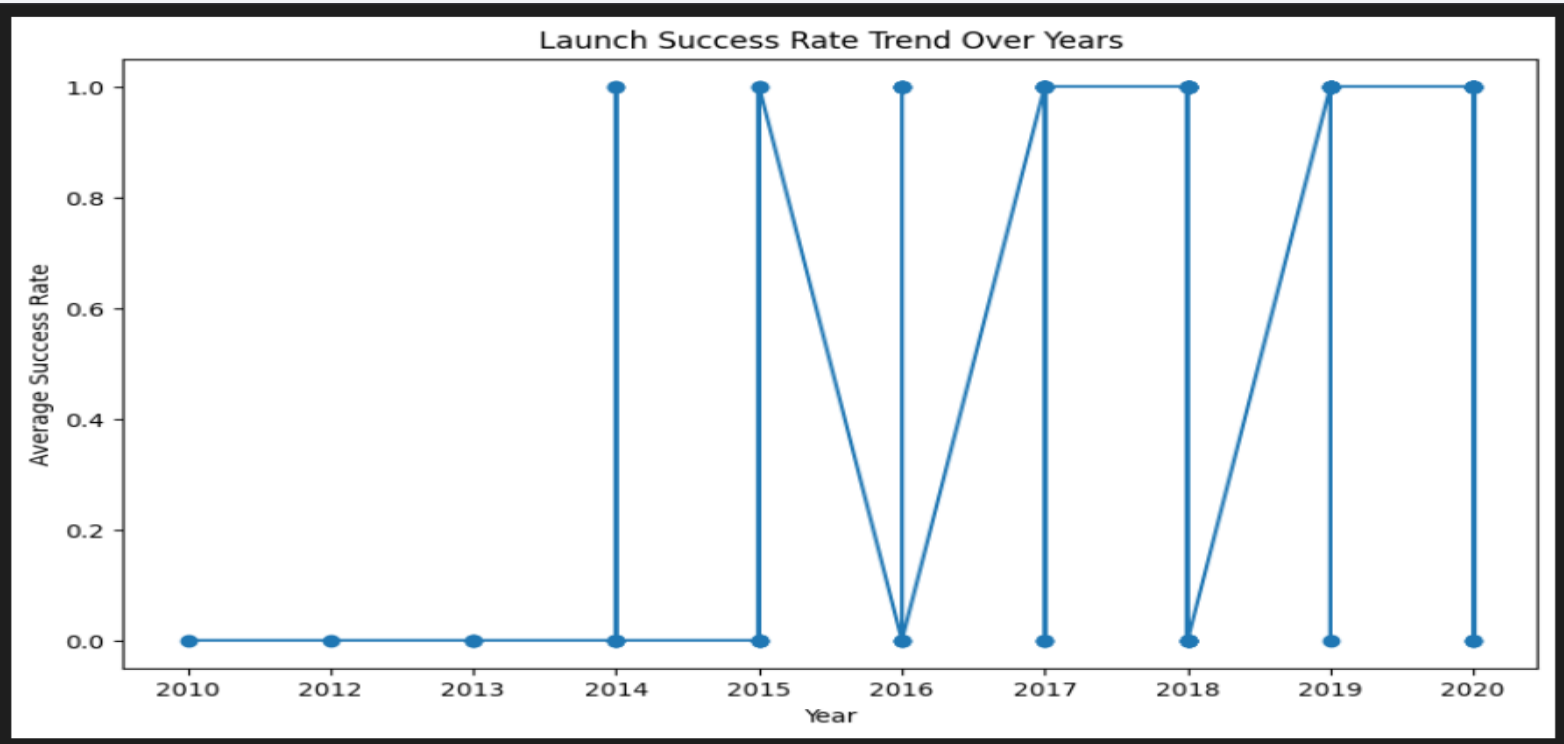
Launch Success Yearly Trend




```
you can observe that the sucess rate since 2013 kept increasing till 2020
Conclusion:
-Launch success rate steadily increases over the years.
-Early Falcon 9 missions experienced more landing failures.
-Recent launches show consistently higher success rates, reflecting improvements in rocket reusability technology.
```

Key Insights from EDA

- Landing success improved significantly over time.
- Payload mass and orbit type influence landing probability.
- LEO missions show the highest landing success rate.
- SpaceX launch sites CCAFS SLC-40 and KSC LC-39A handle the majority of successful launches.

All Launch Site Names

Query Result:

Unique launch sites identified in the dataset:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

-The dataset contains **four primary SpaceX launch sites**.
-Most Falcon 9 launches occurred at **CCAFS SLC-40 and KSC LC-39A**, which are located in Florida, while **VAFB SLC-4E** in California is mainly used for polar orbit missions.

Launch Site Names Begin with 'CCA'

Query Result:

Examples of launch records where the launch site begins with **CCA**:

13]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation:

-The query shows that **Cape Canaveral Air Force Station (CCAFS)** is one of the most frequently used launch locations for Falcon 9 missions.

Total Payload Mass

Query Result:

Total_Payload_Mass
45596

Explanation:

-The result represents the **combined payload weight of all NASA-related launches**.

-This highlights NASA's significant contribution to Falcon 9 missions and demonstrates the rocket's capability to carry large payloads.

Average Payload Mass by F9 v1.1

Query Result

The calculated **average payload mass for booster version F9 v1.1**.

AVERAGE_PAYLOAD_MASS
2928.4

Explanation

-This value shows the **typical payload weight carried by the Falcon 9 v1.1 rocket configuration**.
-It helps compare the payload capacity of different Falcon 9 versions used in various missions.

First Successful Ground Landing Date

Query Result

Date of the **first successful ground landing of a Falcon 9 booster**.

First_Successful_Ground_Pad_Landing
2015-12-22

Explanation

-This mission marked a **major milestone in reusable rocket technology**, demonstrating that Falcon 9 boosters could safely land and be reused in future missions.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query Result

Boosters that successfully landed on a **drone ship** while carrying payloads between **4000 kg and 6000 kg**.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

-The result shows that Falcon 9 boosters can successfully land even when carrying **moderately heavy payloads**, proving the effectiveness of SpaceX's autonomous drone ship landing system.

Total Number of Successful and Failure Mission Outcomes

Query Result

The query counts the number of **successful and failed landing outcomes**.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation

-The results indicate that **successful landings occur more frequently than failures**, reflecting improvements in Falcon 9 landing technology over time.

Boosters Carried Maximum Payload

Query Result

Boosters associated with the **highest payload mass recorded in the dataset**.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Explanation

-These missions represent the **maximum payload capability of Falcon 9 rockets** and typically correspond to more complex missions such as **GTO launches**.

2015 Launch Records

Query Result

List of missions in **2015** where the landing outcome on a **drone ship failed**, including:

- Booster version
- Launch site

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Explanation

-2015 missions show several landing failures as SpaceX was **still developing and testing drone ship landing technology** during early Falcon 9 missions.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query Result

Landing outcomes ranked by frequency between **June 2010 and March 2017**.

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation

-The ranking highlights the **most common landing outcomes**, with successful landings becoming more frequent as SpaceX improved booster recovery techniques.

Key Insights from SQL Analysis

- Falcon 9 launches occur from **four main launch sites**: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.
- Cape Canaveral launch sites account for **a large share of missions**, highlighting Florida as a major launch hub.
- Payload analysis shows that Falcon 9 rockets can carry **very large payload masses**, demonstrating strong launch capability.
- The **first successful ground landing occurred in 2016**, marking an important milestone for reusable rockets.
- Landing success increased over time as **SpaceX improved booster recovery technology**.

Overall Insight:

SQL queries helped extract mission statistics and operational patterns, providing structured insights into **launch sites, payload capacity, and landing outcomes**, which support further data analysis and machine learning modeling.

Section 3

Launch Sites Proximities Analysis

Launch Site Locations on Global Map

Explanation

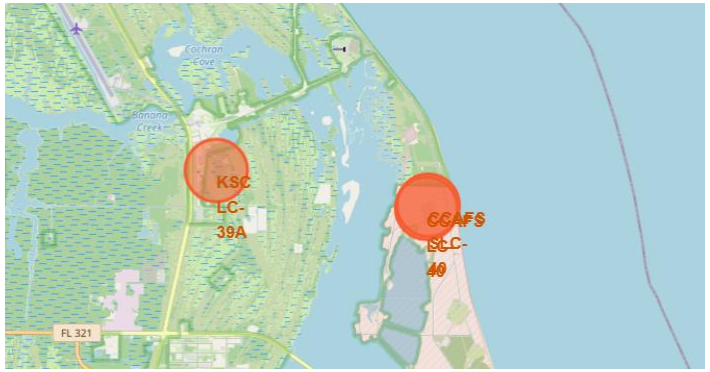
The interactive map displays the geographic locations of all Falcon 9 launch sites.

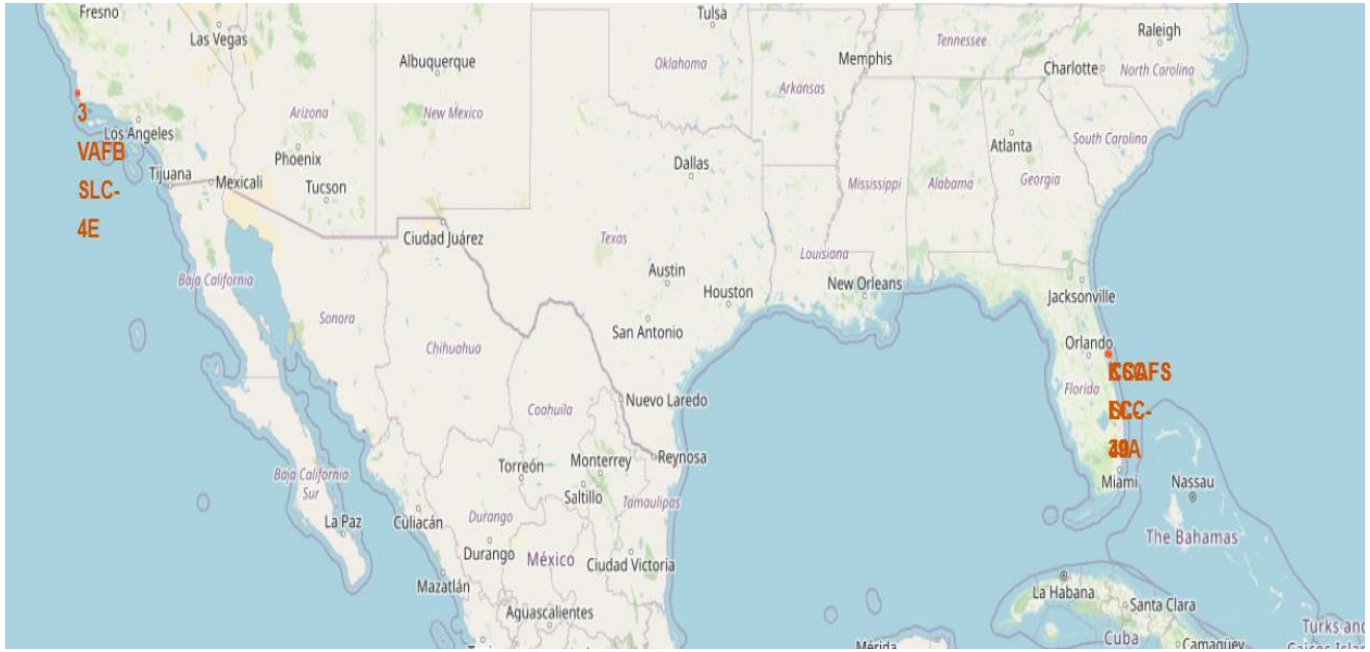
Key elements shown on the map:

- **Markers represent launch site locations**.
- Each marker corresponds to a SpaceX launch facility.
- The sites include **CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E**.

Findings

- Most launch sites are located in **Florida (Cape Canaveral and Kennedy Space Center)**.
- **VAFB SLC-4E in California** is primarily used for polar orbit launches.
- Launch facilities are located **near coastlines**, which improves safety by allowing rockets to launch over the ocean.





Launch Outcomes Visualization on Map

Explanation

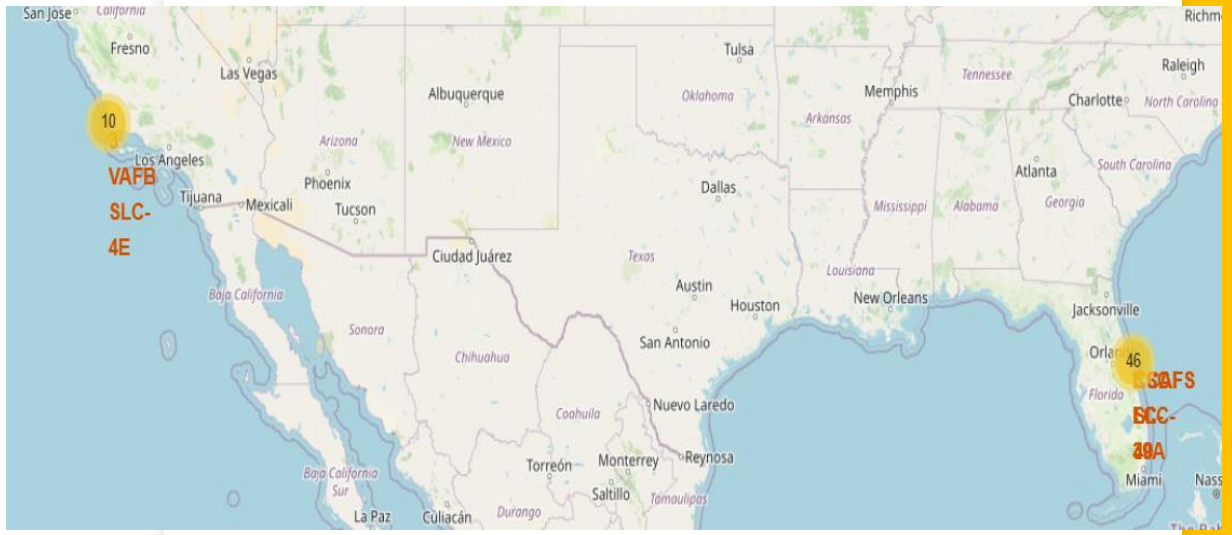
This map visualizes the **launch outcomes using color-coded markers**.

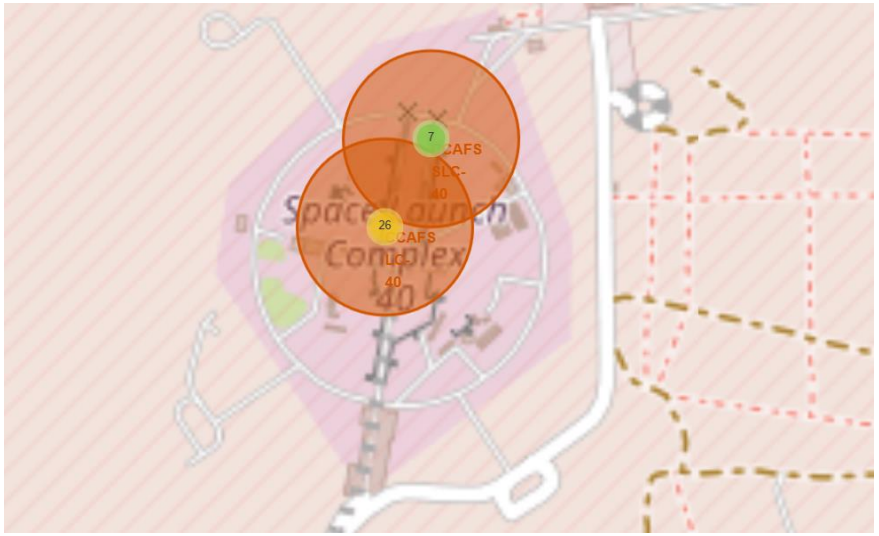
Key elements:

- **Green markers indicate successful landings**.
- **Red markers indicate failed landings**.
- Marker clustering groups nearby launches to improve visualization.

Findings

- A larger number of launches show **successful landing outcomes**, especially in recent missions.
- Launch success is concentrated at **CCAFS SLC-40 and KSC LC-39A**, which handle most Falcon 9 launches.
- The visualization highlights the **improvement in landing success rates over time**.





Launch Site Proximity Analysis

Explanation

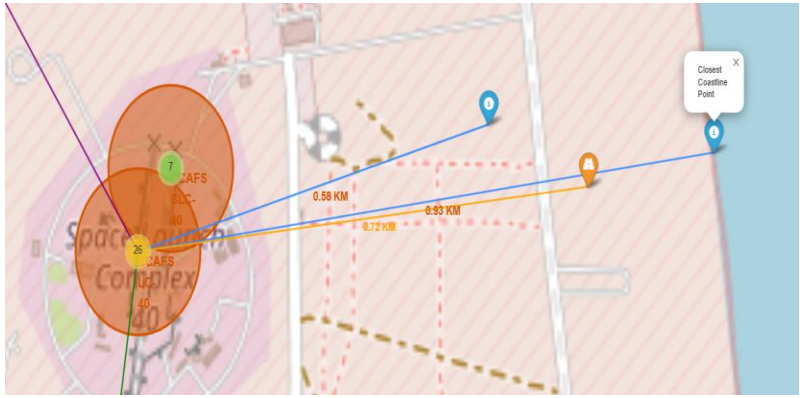
This map shows a **selected launch site and its proximity to surrounding infrastructure**.

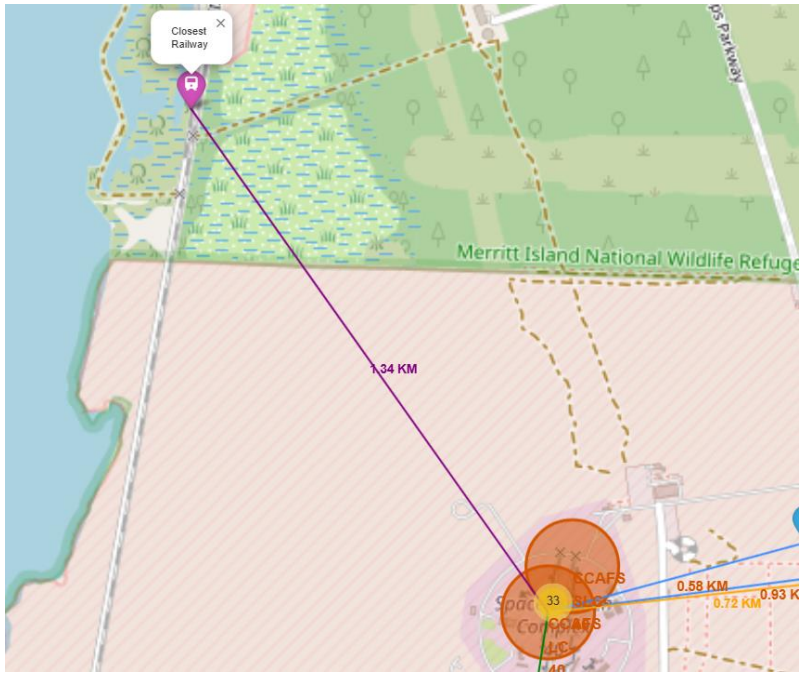
Key elements:

- **Lines display distances between the launch site and nearby features** such as:
  - Railways
  - Highways
  - Coastlines
- Distance values are calculated and displayed directly on the map.

Findings

- Launch sites are located **very close to coastlines**, allowing rockets to launch safely over open water.
- They are typically **far from major cities**, reducing risk to populated areas.
- Some launch sites are located **near transportation infrastructure** such as highways and railways, which helps with logistics and equipment transport.





Folium Map Analysis — Conclusion

Key Geographic Insights:

Analysis of launch site proximity shows clear patterns in how Falcon 9 launch facilities are positioned.

- **Railways:** Launch sites are **not located very close to railways**, indicating that rail transport is not the primary logistical route.
- **Highways:** Launch sites are **relatively close to highways**, allowing convenient transportation of equipment and personnel.
- **Coastlines:** All launch sites are **very close to coastlines**, enabling rockets to launch safely over open ocean areas.
- **Cities:** Launch sites maintain **significant distance from major cities**, reducing risk to populated regions.

Overall Interpretation:

The placement of launch sites is **strategically planned rather than random**.

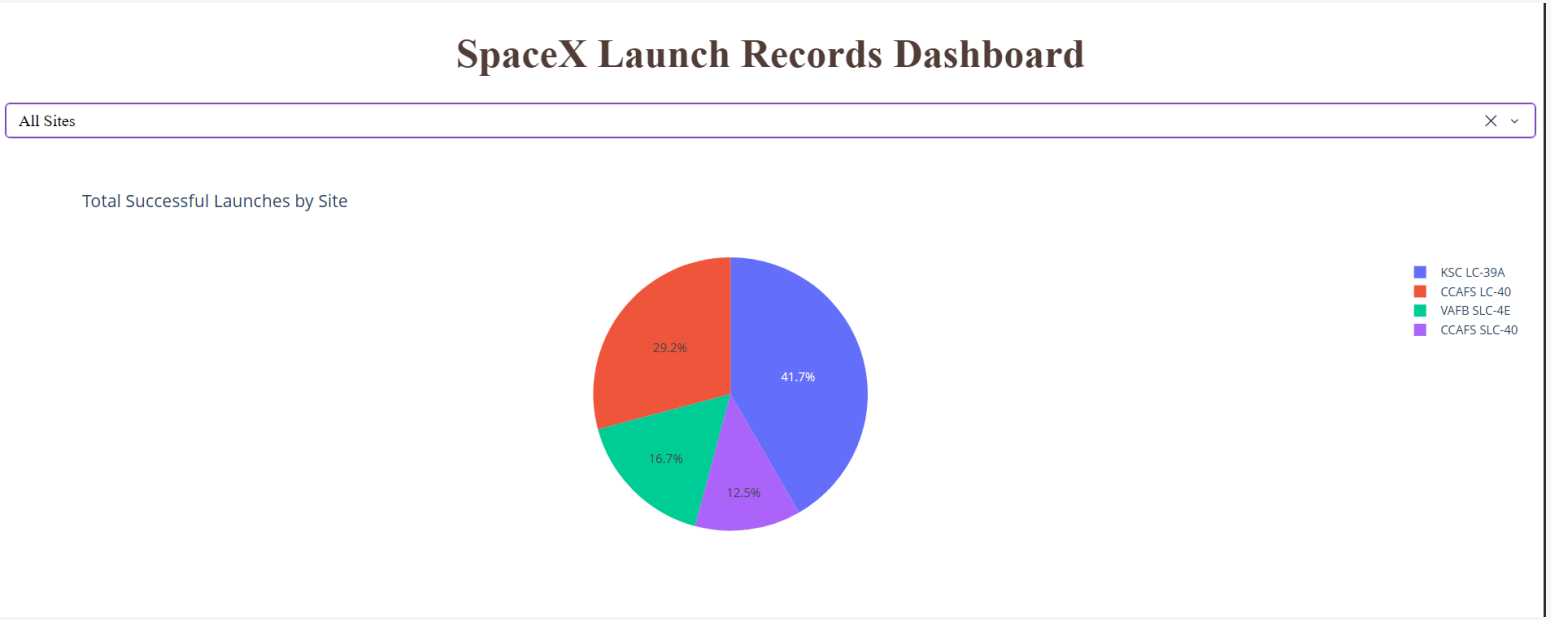
Launch sites are:

- **Intentionally coastal** for safe rocket trajectories
- **Logistically accessible** through nearby highway infrastructure
- **Isolated from dense population centers** for safety
- **Strategically positioned for eastward orbital launches**, benefiting from Earth's rotation

These geographic factors reflect **engineering, safety, and operational priorities in space launch site selection**.

Section 4

Build a Dashboard with Plotly Dash



Launch Success Distribution Across All Sites

Explanation

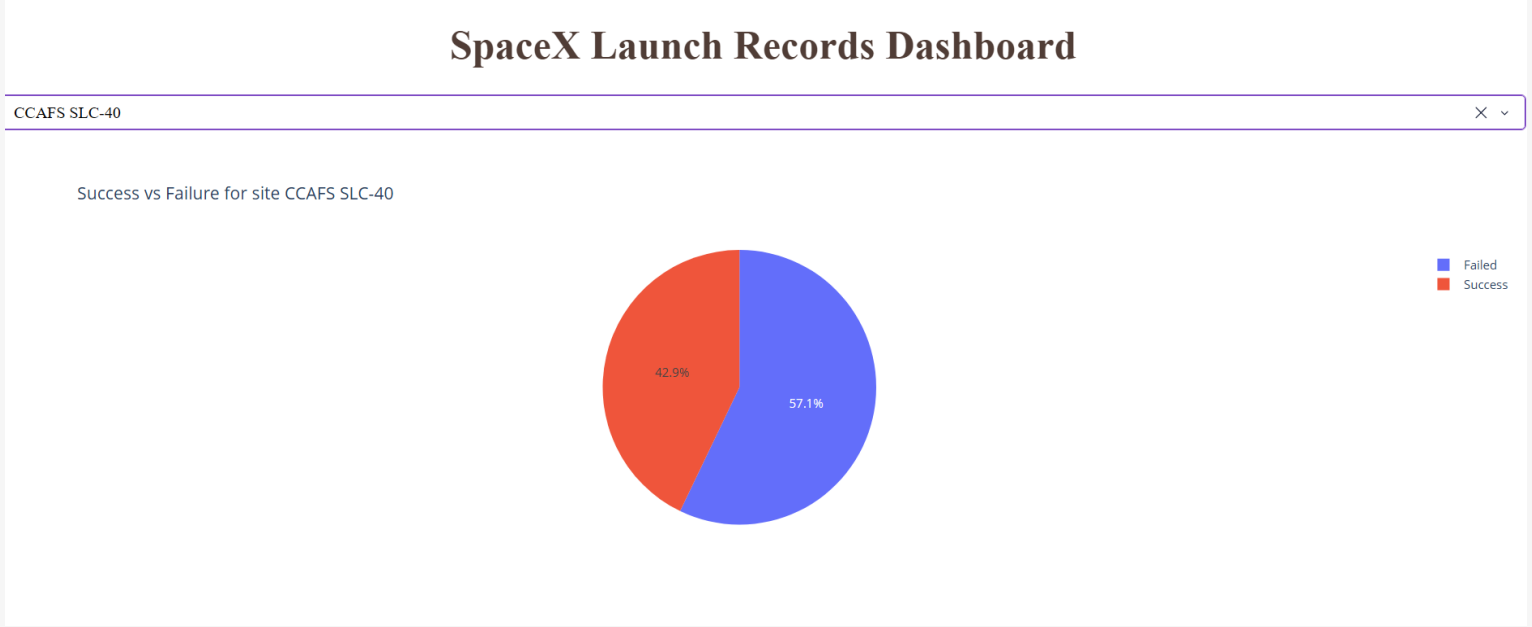
This pie chart shows the **distribution of successful launches across all Falcon 9 launch sites**.

Key elements:

- Each slice represents a **launch site**.
- The size of the slice corresponds to the **number of successful launches**.

Findings:

- **CCAFS LC-40 and KSC LC-39A account for the majority of successful launches**.
- These sites handle most Falcon 9 missions, leading to higher overall success counts.
- Other launch sites contribute fewer missions due to **specific mission types such as polar orbit launches**.



Launch Success Ratio for CCAFS SLC-40

Explanation

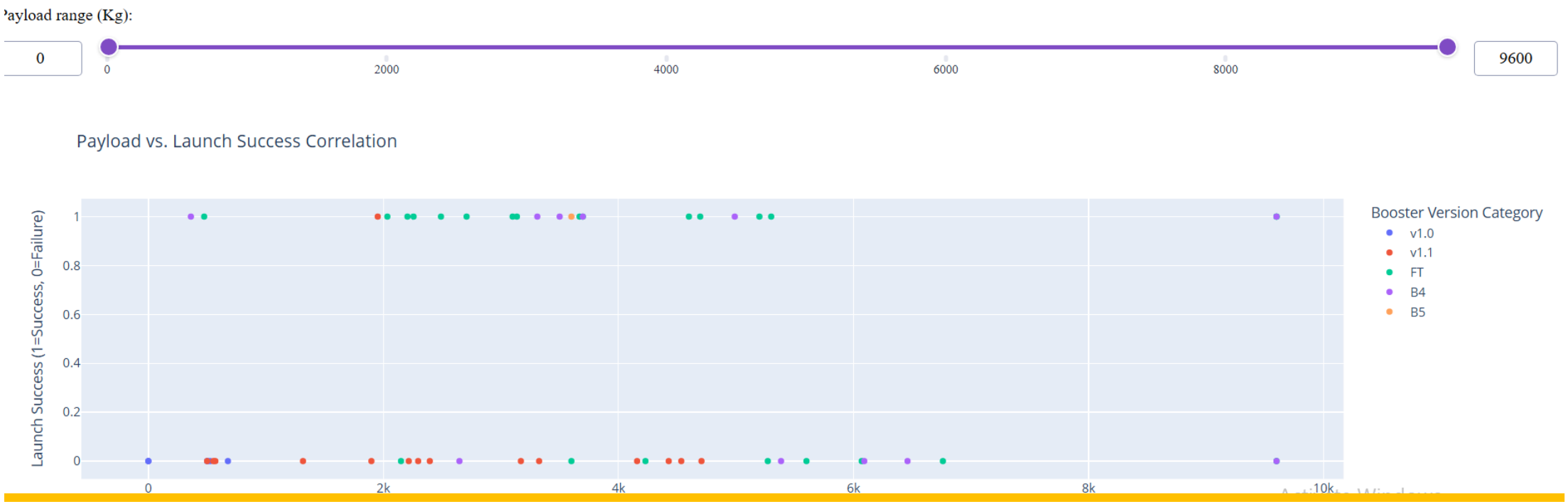
This pie chart shows the **success vs failure ratio for launches at CCAFS SLC-40**, the site with the **highest launch success ratio (42.9%)**.

Key elements:

- The chart separates **successful and failed landing outcomes**.
- The relative sizes of slices represent the **proportion of outcomes**.

Findings:

- CCAFS SLC-40 demonstrates the **highest launch success ratio among all launch sites**.
- This site is heavily used for Falcon 9 missions and benefits from **operational experience and optimized launch infrastructure**.



Payload Mass vs Launch Outcome (All Launch Sites)

Explanation

This scatter plot shows the relationship between **payload mass and launch outcome across all launch sites**.

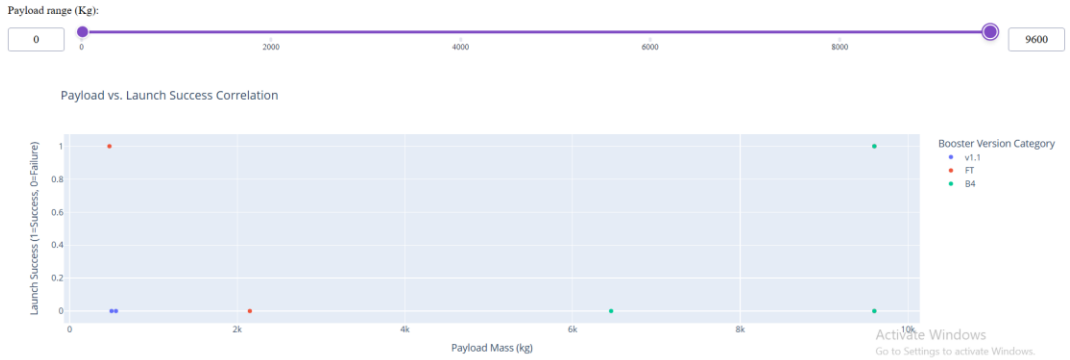
Key elements:

- **X-axis:** Payload Mass (kg)
- **Y-axis:** Launch Outcome (Success or Failure)
- Each point represents an individual launch mission.

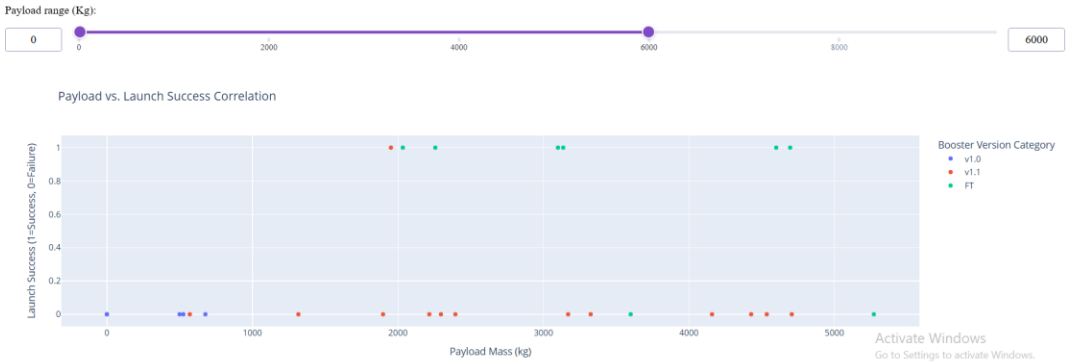
Findings:

- Many successful launches occur within the **moderate payload range (2000–6000 kg)**.
- Early missions with **lower payload masses show more failures**, reflecting early development stages.
- Launch success improves as Falcon 9 technology matured.

VAFB SLC-4E



CCAFS LC-40 :



Payload vs Launch Outcome: CCAFS LC-40 and VAFB SLC-4E

Explanation:

This visualization shows the relationship between **payload mass and launch success** for two launch sites: **CCAFS LC-40** and **VAFB SLC-4E**.

Each point represents a launch mission.
Colors represent **different booster versions**, while the slider above the graph filters the **payload mass range**.

Findings:

CCAFS LC-40

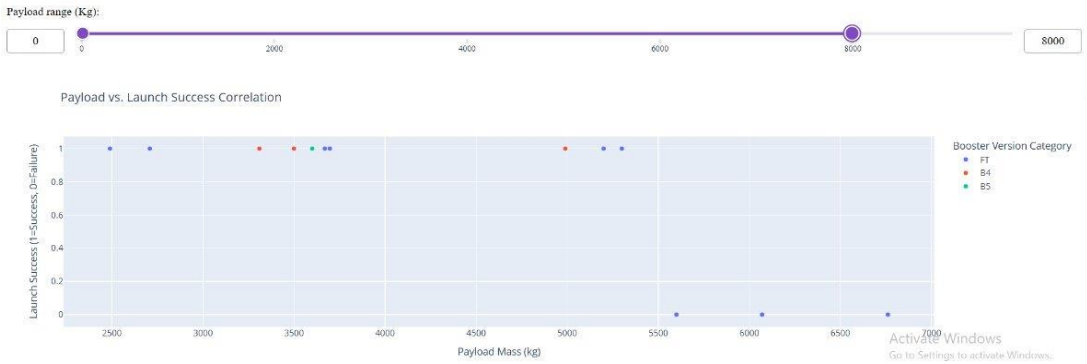
- Most launches occur within **2000–6000 kg payload range**.
- Several successful launches appear in this range, showing stable performance for medium payload missions.
- Some failures occur at lower payload ranges, likely from earlier Falcon 9 missions.

VAFB SLC-4E

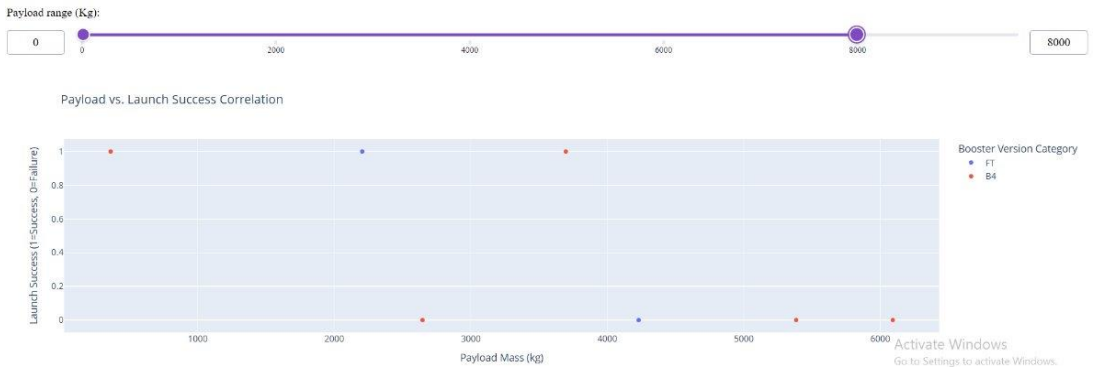
- Fewer launches compared to Florida sites.
- Successful missions occur at **higher payload masses (~8000–9000 kg)**.
- This site is often used for **polar orbit missions**, which explains the different payload distribution.

Payload vs Launch Outcome: KSC LC-39A and CCAFS SLC-40

KSC LC 39-A



CCAFS SLC-40 :



Explanation:

This visualization compares payload mass and landing success for **KSC LC-39A** and **CCAFS SLC-40**, two of the most frequently used Falcon 9 launch sites.

Findings:

KSC LC-39A

- Launches cover a **wide payload range (2000–8000 kg)**.
- Many launches with **medium payload mass show successful landings**.
- This indicates strong reliability for missions launched from Kennedy Space Center.

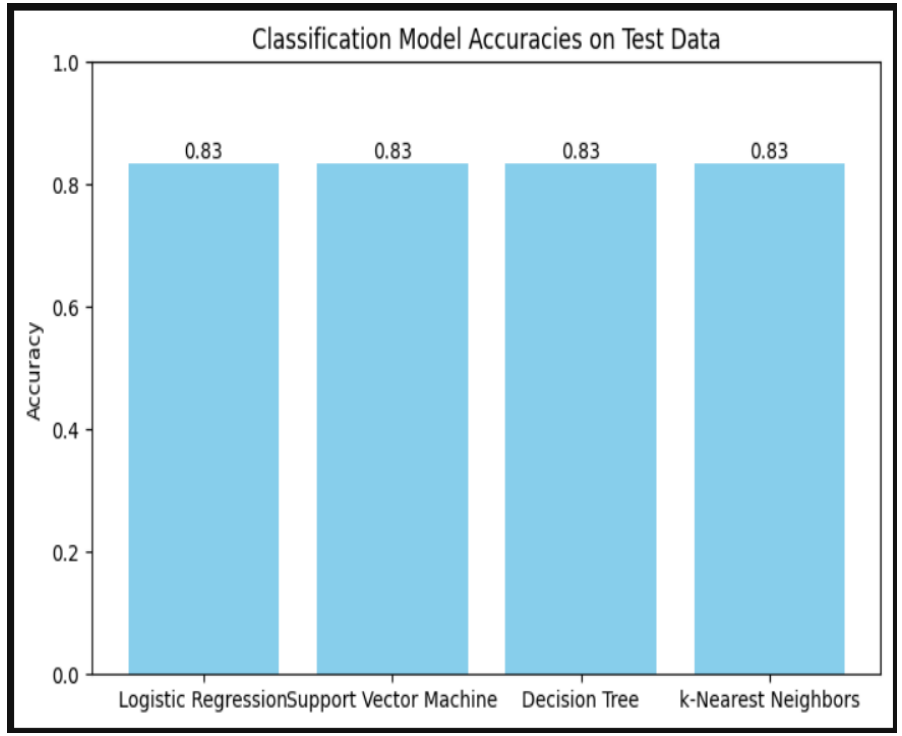
CCAFS SLC-40

- One of the **most active launch sites**.
- Launches mostly occur in the **1000–5000 kg range**.
- Successes are concentrated in the **mid-payload range**, indicating optimized mission configurations.

Section 5

Predictive Analysis (Classification)

Classification Accuracy



- **Four classification models were built and evaluated:**
    - Logistic Regression
    - Support Vector Machine
    - Decision Tree
    - K-Nearest Neighbors
- **Test Accuracy Results**
    - Logistic Regression: **0.83**
    - Support Vector Machine: **0.83**
    - Decision Tree: **0.83**
    - K-Nearest Neighbors: **0.83**
- **Training Accuracy**
    - Logistic Regression: **0.85**
    - Support Vector Machine: **0.85**
    - Decision Tree: **0.88**
    - K-Nearest Neighbors: **0.85**

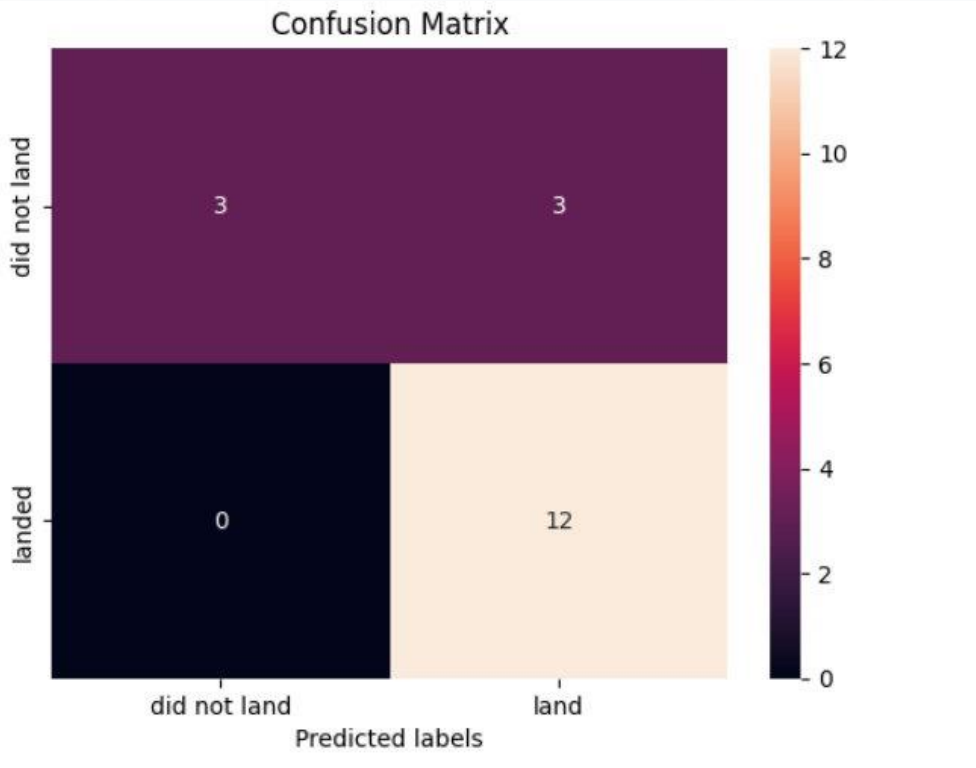
Key Finding:

- All models achieved **similar test accuracy (~83%)**, indicating comparable predictive performance.
- **Decision Tree achieved the highest training accuracy (0.88)**, but this does **not translate into higher test accuracy**, suggesting slight overfitting.
- Since the **test accuracy is identical**, any of the models can reasonably be selected.

Conclusion Statement

The models demonstrate **consistent predictive performance**, suggesting that the dataset features provide stable signals for predicting launch success regardless of algorithm choice.

Confusion Matrix (Decision Tree Classifier)



Interpretation:

- **True Positives (12):** The model correctly predicted 12 successful landings.
- **True Negatives (3):** The model correctly predicted 3 failed landings.
- **False Positives (3):** The model predicted a successful landing when the rocket did not land.
- **False Negatives (0):** The model did not miss any successful landings.

Key Insights

- The model performs **very well at identifying successful landings**.
- There are **no false negatives**, meaning the model never predicts failure when the rocket actually lands.
- Some **false positives exist**, where the model predicts success but the landing fails.

Conclusion:

The model shows **strong performance in detecting successful landings**, which is valuable for launch prediction analysis. However, improving the model's ability to detect failures could further enhance reliability.

Machine Learning Analysis – Key Findings

- **Key Points :**
- Multiple classification models were trained to predict **Falcon 9 first-stage landing success**:
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Tree
  - K-Nearest Neighbors (KNN)
- **Model Performance :**
  - All models achieved **~83% test accuracy**, indicating consistent predictive capability.
  - **Decision Tree** showed slightly higher training accuracy (0.88), suggesting minor overfitting compared to the others.
- **Confusion Matrix Insights :**
  - The models correctly identified **most successful landings**.
  - **No false negatives** were observed, meaning the model did not miss any successful landings.
  - A small number of **false positives** occurred, where the model predicted a landing but the rocket did not land.
  - **Key Takeaway**
  - The models demonstrate that **launch outcomes can be predicted with strong reliability using historical mission data**, showing that key mission parameters contain meaningful predictive patterns.

Project Conclusion – SpaceX Falcon 9 Landing Prediction

"Data analysis and machine learning can reliably predict Falcon 9 landing success."

Data Insights :

- Launch success **improves over time**
- Certain **launch sites perform better**
- Booster reuse impacts outcomes

Geographic Strategy:

- Launch sites **near coastlines**
- **Accessible for logistics**
- **Far from major cities**
- Positioned for **efficient eastward launches**

Machine Learning Results:

- 4 models trained
- **~83% prediction accuracy**
- Strong detection of **successful landings**

Data-driven analysis helps understand launch success patterns and supports future mission planning.

Appendix

Data Collection: Used SpaceX REST API for data collection

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

```
response = requests.get(spacex_url)
```

Data Visualization: Visualized flight numbers per launch site during EDA

```
sns.catplot(x='FlightNumber', y='LaunchSite', hue='Class', data=df, aspect=5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Launch Site",fontsize=20)
plt.show()
```

SQL Queries: Used to calculate average payload mass of booster version F9 v1.1

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") AS AVERAGE_PAYLOAD_MASS FROM SPACEXTABLE WHERE "Booster_Version"='F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

AVERAGE_PAYLOAD_MASS
2928.4

Data Preparation: Prepared dataset for training and evaluating machine learning models.

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)
```

Machine Learning Model: Trained classification models to predict Falcon 9 first-stage landing success.

```
parameters ={"C":[0.01,0.1,1],'penalty':['l2'], 'solver':['lbfgs']}# l1 lasso l2 ridge
lr=LogisticRegression()
logreg_cv = GridSearchCV(lr, parameters, cv=10)
logreg_cv.fit(X_train, Y_train)
```

Thank you!